# Resit programming with Python

**Deadline: 12/03/2026, 23:59, please submit your *.py or *.ipynb files using the designated folder on the K2 site of the course.**

## Project 1

The task is to predict house prices in different communities based on air pollution level (represented by the concentration of nitrogine oxid), distance from employment centers, average student-teacher ratio in local schools and average number of rooms, and crime rate The variables are as follows:

- *price*
  median housing price
- *crime*
  crimes committed per capita
- *nox*
  nitrogine oxide, parts per 100 mill.
- *rooms*
  avg number of rooms per house
- *dist*
  weighted dist. to 5 employ centers
- *stratio*
  average student-teacher ratio
- *lprice*
  log(*price*)
- *lnox*
  log(*nox*)

The .csv file can be found at `https://kaggle.com/datasets/f06415afa98ac5a0a50ad7fdec3c4b5260 23dcf2a1b39435e6f2b6a43eddf88e` or downloaded from the assignment in K2 (`hprice2_merged.csv`)

Use various models (linear regression, SVM, decision tress, neural networks) for predicting house prices. Compare the various models in terms of accuracy of prediction. Which one is the best ? Select one or two independent variables and use them to build models for predicting houses prices. Which variables give the best model for predicting house prices ? Try to predict the logarithm of house prices instead of house prices. Do you see any improvement in performance of the models ?